

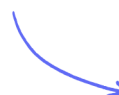
Multiple Choice Questions

Groups 1 & 2

Ionisation Energy - Groups 1 & 2 / Reactivity - Groups 1 & 2 / Oxides & Hydroxides / Group 2 Hydroxides & Sulfates / Nitrates & Carbonates / Flame Tests / Qualitative Tests

Easy (1 question)	/1
Medium (3 questions)	/3
Total Marks	/4

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Easy Questions

1 Which of these compounds does **not** produce a colour in a flame test, and produces an alkaline gas when warmed with sodium hydroxide solution?

A. $\text{Ca}(\text{OH})_2$

B. $\text{Mg}(\text{OH})_2$

C. NH_4Cl

D. BeCl_2

(1 mark)

Medium Questions

1 What is the ionic equation for the reaction between aqueous solutions of barium nitrate and sodium sulfate?

- A. $\text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NaNO}_3(\text{s})$
- B. $2\text{Ba}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Ba}_2\text{SO}_4(\text{s})$
- C. $\text{Na}^{2+}(\text{aq}) + 2\text{NO}_3^-(\text{aq}) \rightarrow \text{Na}(\text{NO}_3)_2(\text{s})$
- D. $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$

(1 mark)

2 Data for some Group 1 and Group 2 cations are shown in the table.

Cation	Ionic radius / nm	Ionic charge
W	0.100	+2
X	0.138	+1
Y	0.113	+2
Z	0.149	+1

Which cation would be expected to form the nitrate with the **greatest** thermal stability?

- A. W
- B. X
- C. Y
- D. Z

(1 mark)

3 Magnesium nitrate decomposes on heating.



What is the total volume of gas formed at room temperature and pressure (r.t.p.) when 0.005 mol of magnesium nitrate decomposes completely?

[Molar volume of a gas at r.t.p. = 24 000 cm³ mol⁻¹]

A. 600 cm³

B. 300 cm³

C. 240 cm³

D. 120 cm³

(1 mark)